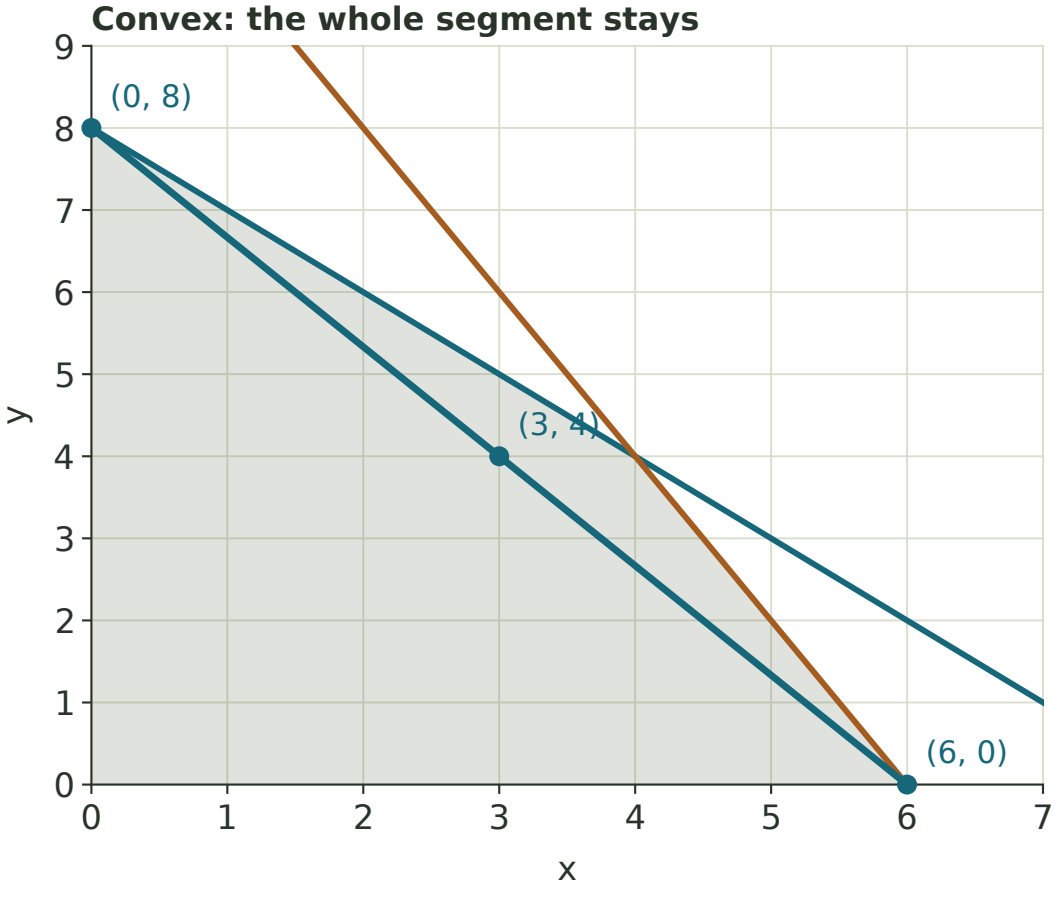
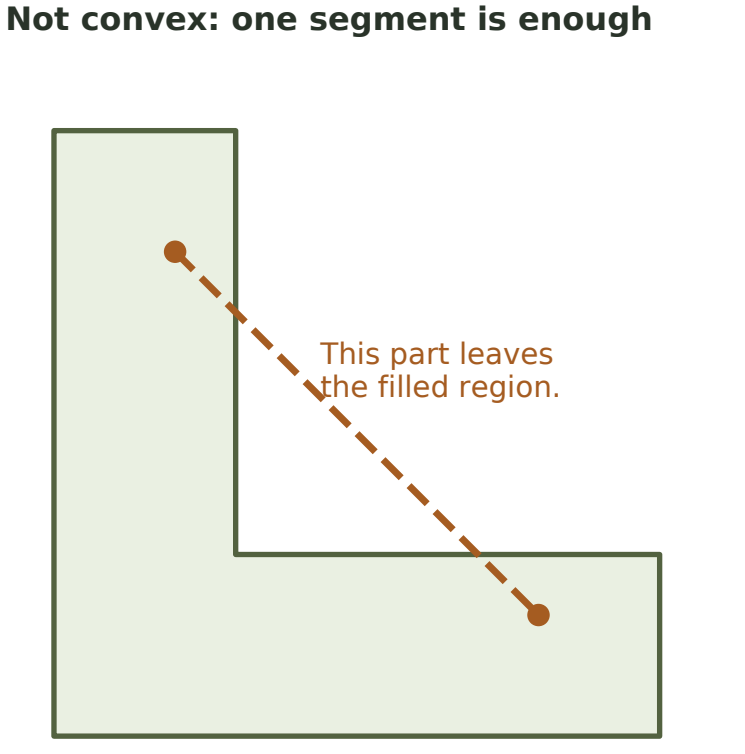

Convex: the whole segment stays
(0, 8)
(3, 4)
(6, 0)
x
y
0
1
2
3
4
5
6
7
8
9


Not convex: one segment is enough
This part leaves
the filled region.